

FPGA PROJECT (EC9039)

CORDIC ALGORITHM

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INTRODUCTION

In the realm of digital signal processing, the Coordinate Rotation Digital Computer (CORDIC) algorithm stands as a versatile and efficient method for trigonometric computations. In this project, we delve into the design and simulation of a CORDIC module using Verilog. Our endeavor is to encapsulate the essence of CORDIC's iterative approach, focusing on precision and efficiency.

THEORY AND CONCEPTS

Coordinate Rotation Digital Computer (CORDIC) Algorithm:

Summary: The COordinate Rotation DIgital Computer (CORDIC) is a specialized digital computing technique designed for real-time applications, especially in airborne computation. It utilizes a unique method, particularly suitable for solving trigonometric relationships in plane coordinate rotation and conversion from rectangular to polar coordinates. CORDIC, being an entire-transfer computer, consists of a special serial arithmetic unit with shift registers, adder-subtractors, and specialized interconnections. It can solve equations such as:

$$X' = K(Y \cos \theta + X \sin \theta) \quad X = K(Y \cos \theta + X \sin \theta)$$

$$Y' = K(X \cos \theta - Y \sin \theta) \quad Y = K(X \cos \theta - Y \sin \theta)$$

Here, K is a constant, and the angles θ are controlled by a prescribed sequence of conditional additions or subtractions.

THEORY AND CONCEPTS

Introduction: The CORDIC computing technique was developed for real-time digital computers that require frequent computation of trigonometric relationships, especially in navigation equations and coordinate transformations. The CORDIC I prototype was designed based on this technique, providing a high solution rate for such trigonometric relationships.

Functional Description:

CORDIC operations can be functionally described as the digital equivalent of an analog resolver, with two primary modes:

- **Rotation Mode:**
 - Involves computing the new coordinate components after rotating the original vector by a given angle.
- **Vectoring Mode:**
 - Computes the magnitude and angle of the original vector from its coordinate components.

ROTATIONAL MODE

- ROTATE AN INITIAL VECTOR (x_{in}, y_{in}) BY θ
- DECOMPOSE THE ANGLE

$$z[j + 1] = z[j] - \sigma_j \tan^{-1}(2^{-j})$$

$$z[0] = \theta \quad x[0] = x_{in} \quad y[0] = y_{in}$$

$$\sigma_j = \begin{cases} 1 & \text{if } z[j] \geq 0 \\ -1 & \text{if } z[j] < 0 \end{cases}$$

- PERFORM MICRO-ROTATIONS
- FINAL VALUES

$$x_f = K(x_{in} \cos \theta - y_{in} \sin \theta)$$

$$y_f = K(x_{in} \sin \theta + y_{in} \cos \theta)$$

$$z_f = 0$$

Micro rotations in rotational mode

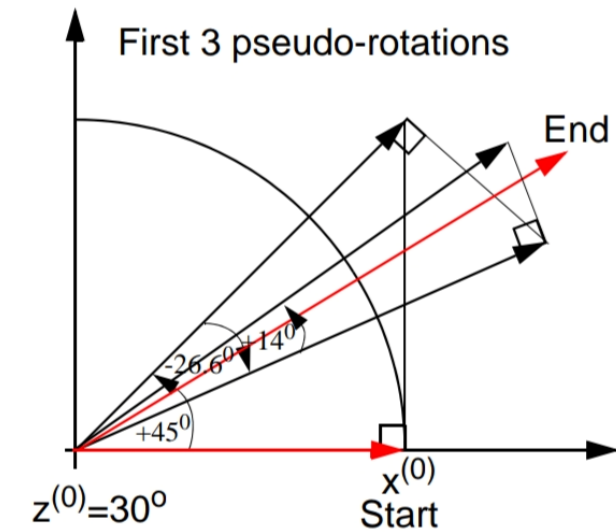
In Rotation Mode the decision operator d_i obeys the condition:

$$d_i = \text{sign}(z^{(i)})$$

Thus, we input $x^{(0)}$ and $z^{(0)}$ ($y^{(0)} = 0$) and then drive $z^{(0)}$ towards 0.

Example: calculate $\sin z^{(0)}$, $\cos z^{(0)}$ where $z^{(0)} = 30^\circ$

i	d_i	$\theta^{(i)}$	$z^{(i)}$	$y^{(i)}$	$x^{(i)}$
0	+1	45	+30	0	0.6073
1	-1	26.6	-15	0.6073	0.6073
2	+1	14	+11.6	0.3036	0.9109
3	-1	7.1	-2.4	0.5313	0.8350
4	+1	3.6	+4.7	0.4270	0.9014
5	+1	1.8	+1.1	0.4833	0.8747
6	-1	0.9	-0.7	0.5106	0.8596
7	+1	0.4	+0.2	0.4972	0.8676
8	-1	0.2	-0.2	0.5040	0.8637
9	+1	0.1	+0	0.5006	0.8657



VECTOR MODE

- ROTATE INITIAL VECTOR (x_{in}, y_{in}) UNTIL $y = 0$
- FOR INITIAL VECTOR IN THE FIRST QUADRANT:

$$\sigma_j = \begin{cases} 1 & \text{if } y[j] < 0 \\ -1 & \text{if } y[j] \geq 0 \end{cases}$$

- ACCUMULATE ROTATION ANGLE IN z
- FOR $x[0] = x_{in}$, $y[0] = y_{in}$ and $z[0] = z_{in}$, THE FINAL VALUES ARE

$$x_f = K(x_{in}^2 + y_{in}^2)^{1/2}$$

$$y_f = 0$$

$$z_f = z_{in} + \tan^{-1}\left(\frac{y_{in}}{x_{in}}\right)$$

Micro rotations in vector mode

Notes:

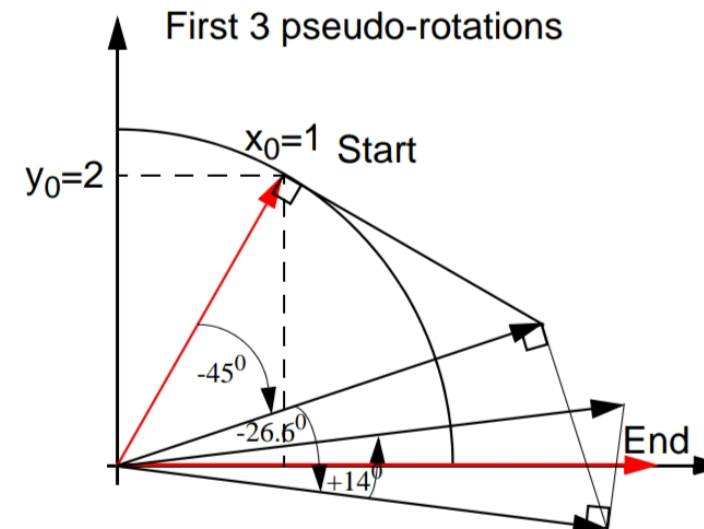
In Vectoring Mode the decision operator d_i obeys the condition:

$$d_i = -\text{sign}(x^{(i)}y^{(i)})$$

Thus, we input $x^{(0)}$ and $y^{(0)}$ ($z^{(0)} = 0$) and then drive $y^{(0)}$ towards 0.

Example: calculate $\tan^{-1}(y^{(0)}/x^{(0)})$ where $y^{(0)} = 2$ and $x^{(0)} = 1$

i	$z^{(i)}$	θ^i	$y^{(i)}$
0	0	45	2
1	45	26.6	1
2	71.6	14	-0.5
3	57.6	7.1	0.375
4	64.7	3.6	-0.078
5	61.1	1.8	0.151
6	62.9	0.9	0.039
7	63.8	0.4	-0.019
8	63.4	0.2	0.009



UNIFIED DESCRIPTION

- $m = 1$ FOR CIRCULAR COORDINATES
- $m = -1$ FOR HYPERBOLIC COORDINATES
- $m = 0$ FOR LINEAR COORDINATES
- UNIFIED MICROROTATION IS

$$\begin{aligned}
 x[j+1] &= x[j] - m\sigma_j 2^{-j} y[j] \\
 y[j+1] &= y[j] + \sigma_j 2^{-j} x[j] \\
 z[j+1] &= \begin{cases} z[j] - \sigma_j \tan^{-1}(2^{-j}) & \text{if } m = 1 \\ z[j] - \sigma_j \tanh^{-1}(2^{-j}) & \text{if } m = -1 \\ z[j] - \sigma_j (2^{-j}) & \text{if } m = 0 \end{cases}
 \end{aligned}$$

ALSO $z[j+1] = z[j] - \sigma_j m^{-1/2} \tan^{-1}(m^{1/2} 2^{-j})$

- THE SCALING FACTOR IS

$$K_m[j] = (1 + m 2^{-2j})^{1/2}$$

CORDIC MODES

Table 11.5: UNIFIED CORDIC

Coordinates	Rotation mode $\sigma_j = \text{sign}(z[j])^+$	Vectoring mode $\sigma_j = -\text{sign}(y[j])^+$
Circular ($m = 1$) $\alpha_j = \tan^{-1}(2^{-j})$ initial $j = 0$ $j = 0, 1, 2, \dots, n$ $K_1 \approx 1.64676$ $\theta_{max} \approx 1.74329$	$x_f = K_1(x_{in} \cos(z_{in}) - y_{in} \sin(z_{in}))$ $y_f = K_1(x_{in} \sin(z_{in}) + y_{in} \cos(z_{in}))$ $z_f = 0$	$x_f = K_1(x_{in}^2 + y_{in}^2)^{1/2}$ $y_f = 0$ $z_f = z_{in} + \tan^{-1}(\frac{y_{in}}{x_{in}})$
Linear ($m = 0$) $\alpha_j = 2^{-j}$ initial $j = 0$ $j = 0, 1, 2, \dots, n$ $K_0 = 1$ $\theta_{max} = 2 - 2^{-n}$	$x_f = x_{in}$ $y_f = y_{in} + x_{in}z_{in}$ $z_f = 0$	$x_f = x_{in}$ $y_f = 0$ $z_f = z_{in} + \frac{y_{in}}{x_{in}}$
Hyperbolic ($m = -1$) $\alpha_j = \tanh^{-1}(2^{-j})$ initial $j = 1$ $j = 1, 2, 3, 4, 5 \dots 13, 13, \dots$ $K_{-1} \approx 0.82816$ $\theta_{max} \approx 1.11817$	$x_f = K_{-1}(x_{in} \cosh(z_{in}) + y_{in} \sinh(z_{in}))$ $y_f = K_{-1}(x_{in} \sinh(z_{in}) + y_{in} \cosh(z_{in}))$ $z_f = 0$	$x_f = K_{-1}(x_{in}^2 - y_{in}^2)^{1/2}$ $y_f = 0$ $z_f = z_{in} + \tanh^{-1}(\frac{y_{in}}{x_{in}})$

⁺ $\text{sign}(a) = 1$ if $a \geq 0$, $\text{sign}(a) = -1$ if $a < 0$.

OTHER FUNCTIONS

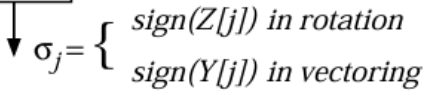
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Table 11.6: SOME ADDITIONAL FUNCTIONS

m	Mode	Initial values			Functions	
		x_{in}	y_{in}	z_{in}	x_R	y_R or z_R
1	rotation	1	0	θ	$\cos \theta$	$y_R = \sin \theta$
-1	rotation	1	0	θ	$\cosh \theta$	$y_R = \sinh \theta$
-1	rotation	a	a	θ	ae^θ	$y_R = ae^\theta$
1	vectoring	1	a	$\pi/2$	$\sqrt{a^2 + 1}$	$z_R = \cot^{-1}(a)$
-1	vectoring	a	1	0	$\sqrt{a^2 - 1}$	$z_R = \coth^{-1}(a)$
-1	vectoring	$a + 1$	$a - 1$	0	$2\sqrt{a}$	$z_R = 0.5 \ln(a)$
-1	vectoring	$a + \frac{1}{4}$	$a - \frac{1}{4}$	0	$\sqrt{a}$	$z_R = \ln(\frac{1}{4}a)$
-1	vectoring	$a + b$	$a - b$	0	$2\sqrt{ab}$	$z_R = 0.5 \ln(\frac{a}{b})$

Note: the final values x_R and y_R are obtained after compensation of the scale factor.

IMPLEMENTATIONS



Computation of Sine and cosine using Cordic rotational mode

Using the circular coordinates (CC) system CORDIC algorithm with the input vector $(1, 0)$ and angle θ we get the following output:

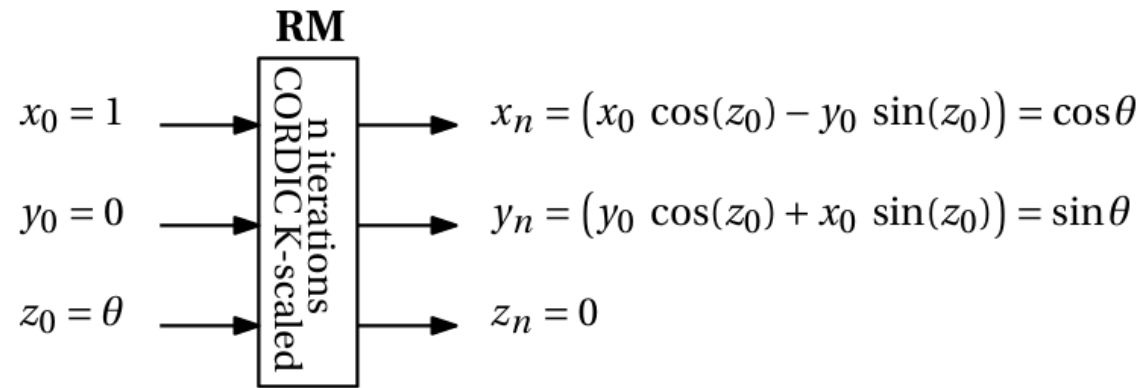
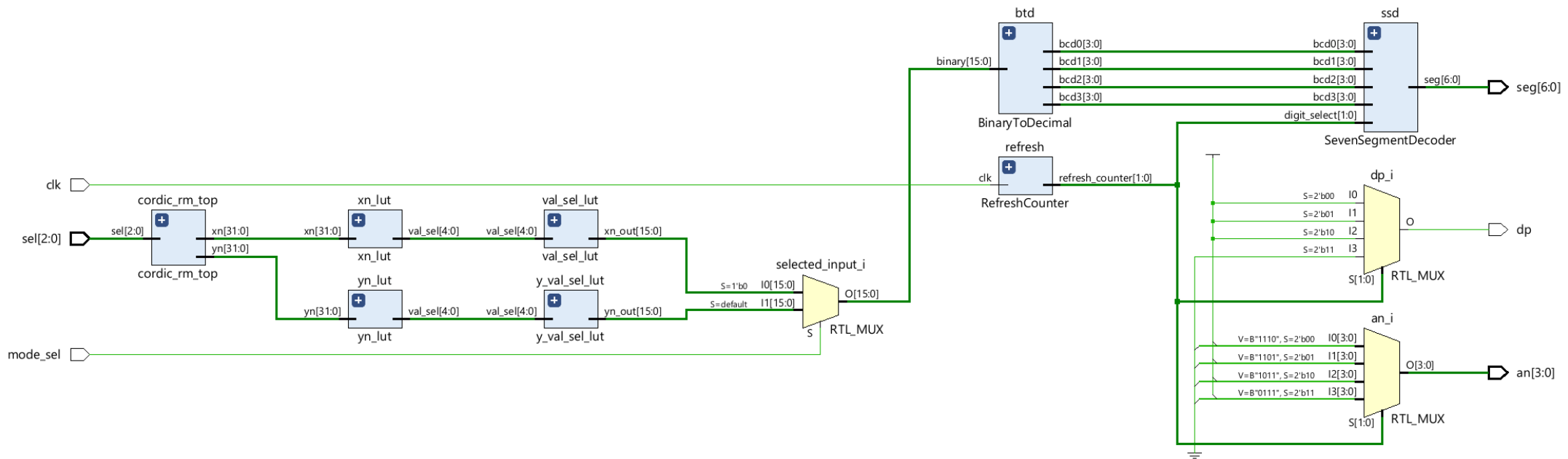


Figure 3.2: *CC-RM Cordic to compute $\sin \theta$, $\cos \theta$ and $\tan \theta$*

It directly computes $\sin \theta$ and $\cos \theta$; $\tan \theta$ can then be derived.

Datapath for Cordic in rotational mode

Application – Sine and cosine functions



Multiplication and division using LC-CORDIC

Using the linear coordinates (LC) system CORDIC algorithm multiplications and division could also be computed.

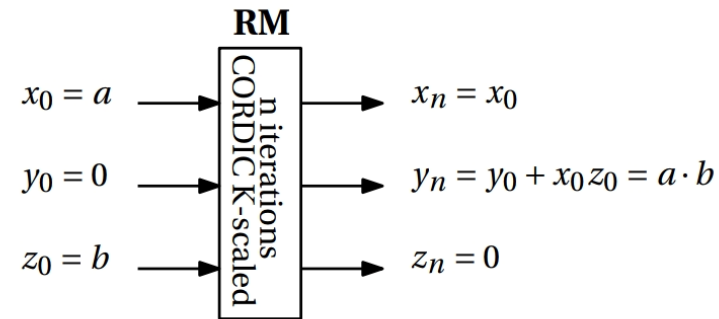


Figure 3.8: *LC-RM Cordic to compute $a \cdot b$*

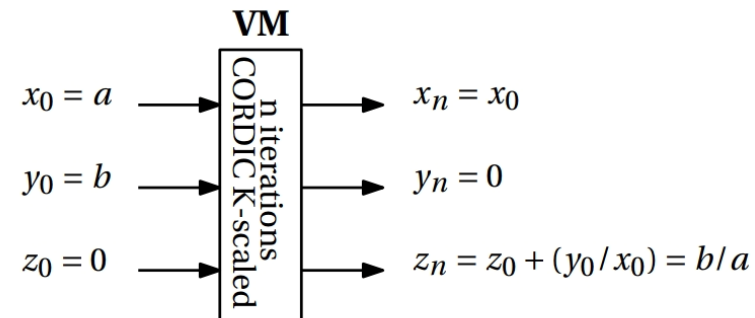
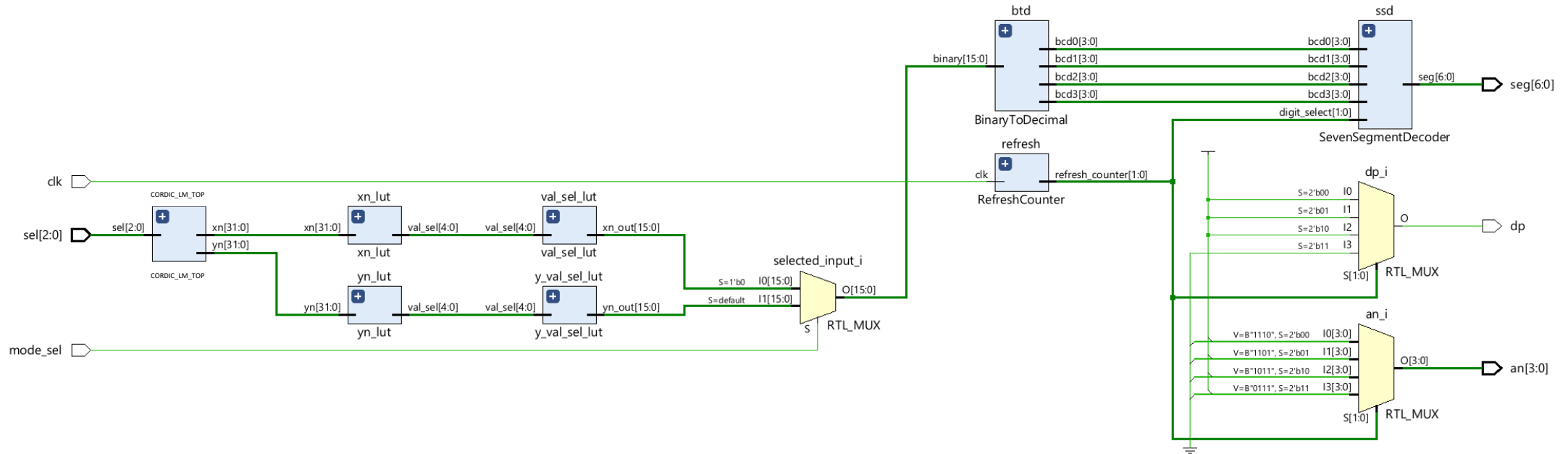


Figure 3.9: *LC-VM Cordic to compute b/a*

Datapath for Cordic in linear mode

Application – Multiplication and division functions



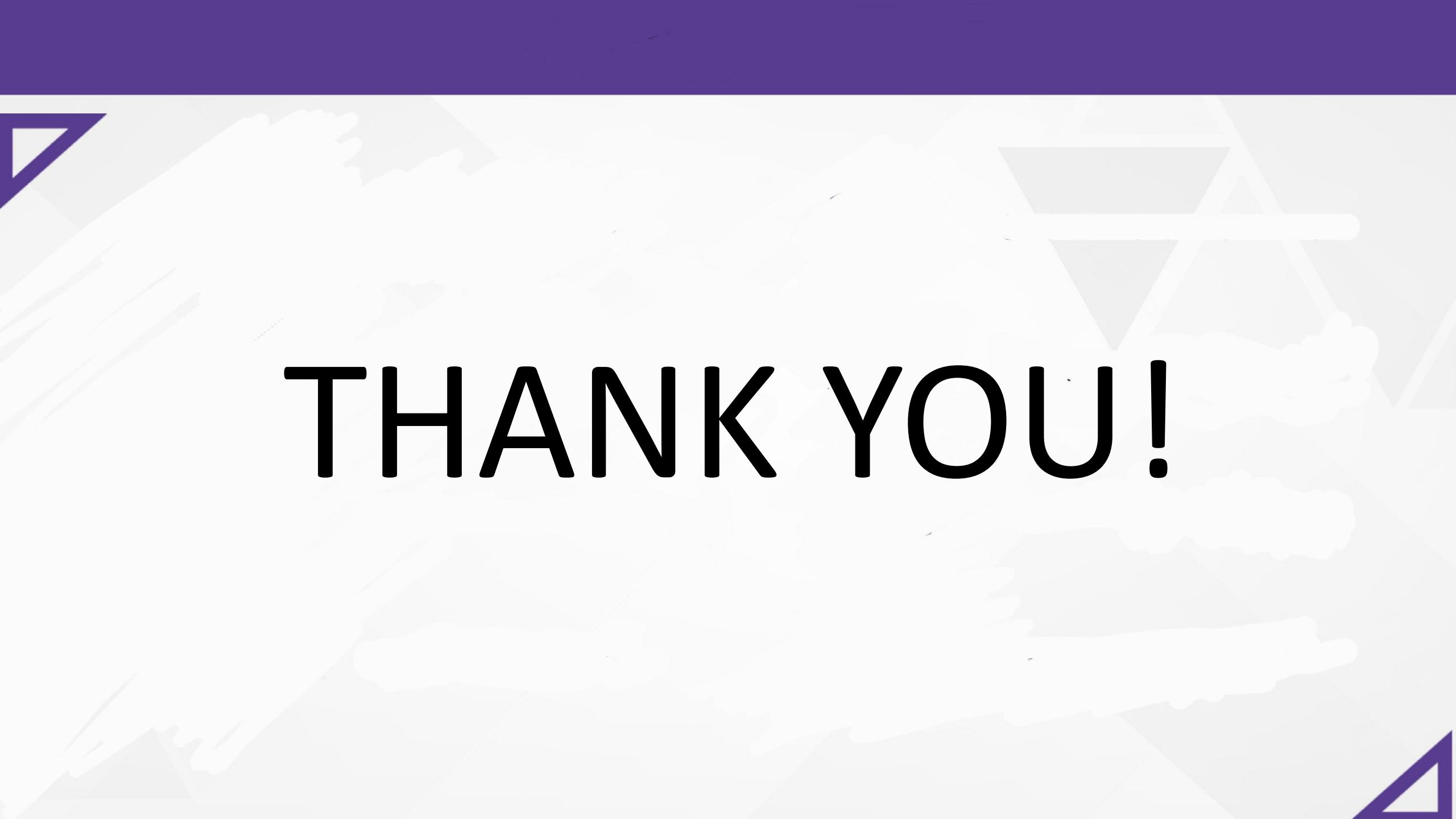
Future scopes

- **DSP:** High-speed filtering, modulation, and PLL.
- **Embedded Systems:** Low-power real-time computation in IoT and robotics.
- **Hardware Efficiency:** Energy-saving, adaptive precision accelerators.
- **Pipelining:** High-throughput CORDIC architectures for parallel processing.
- **Biomedical:** Real-time bio-signal processing and medical imaging.
- **Communications:** MIMO signal rotation, beamforming, and phase calculation.
- **Machine Learning:** Efficient activation functions and coordinate transformations.
- **Navigation:** Real-time positioning and orientation in satellites and vehicles.

- REALIZATION OF ROTATIONS
- CALCULATION OF TRIGONOMETRIC FUNCTIONS
- CALCULATION OF INVERSE TRIGONOMETRIC FUNCTION $\tan^{-1}(a/b)$
- CALCULATION OF $\sqrt{a^2 + b^2}$, etc.
- EXTENDED TO HYPERBOLIC FUNCTIONS
- DIVISION AND MULTIPLICATION
- CALCULATION OF SQRT, LOG, AND EXP
- FOR LINEAR TRANSFORMS, DIGITAL FILTERS, AND SOLVING LIN. SYSTEMS
- MAIN APPLICATIONS: DSP, IMAGE PROCESSING, 3D GRAPHICS, ROBOTICS.

CONCLUSION

In summary, our venture into designing and simulating the CORDIC algorithm in Verilog underscores the fusion of precision, efficiency, and versatility in digital signal processing. By adhering to established principles and leveraging innovative techniques such as pipelining, we achieve robust computational capabilities suitable for a myriad of applications.

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THANK YOU!